

	velocity $\propto \sqrt{1/(1-v^2/c^2)}$ correct
6.	Ans: C

$$\tan(3.8^\circ) = \frac{\text{centripetal acceleration}}{9.81}$$

$$\text{centripetal acceleration} = 0.65158 \text{ m s}^{-2}$$

$$\frac{v^2}{r} = 0.65158 \text{ m s}^{-2}$$

$$r = \frac{\left(\frac{57.8}{3.6}\right)^2}{0.65158} = 396 \text{ m}$$